

Wooah Choi

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MSc AI at Queen Mary University of London, dissertation applying mechanistic interpretability to a PPO-trained causal Transformer in partially observable RL: norm-preserving source exchange reversed both branch decisions while padding exchange reversed neither; in a preregistered fixed-length retrieval follow-up the Transformer held 1.000 accuracy through 32 bindings while matched and larger LSTMs fell to 0.77 or below at eight, in 6/6 seeds. Former Software Engineer at Konan Technology, cutting Supreme Court litigation search latency from 5+ minutes to 8 seconds. Co-founder of GridFlow Trade (QMUL QIncubator), UK energy price forecasting pipeline achieving MAE £4.10/MWh ($R^2=0.943$), 76% error reduction. Experienced across the full ML stack: NLP/BERT, XGBoost, LSTM, Transformer architectures, and PyTorch-based RL.

EDUCATION

MSc in Artificial Intelligence | Queen Mary University of London, London, UK (Sep 2025 - Present)

- Dissertation: Causal Cue Retrieval and Greedy Route Execution: A Mechanistic Analysis of a Transformer Policy in a Fixed MiniGrid T-Maze
- Core Modules: Machine Learning, Reinforcement Learning, Neural Networks and NLP, Information Retrieval, Conversational Agents and Dialogue Systems

Bachelor of Science in Computer Science (Double Major: Statistics and Data Science) | Korea National Open University, Seoul, South Korea (Mar 2021 - Aug 2023)

- Core Modules: Discrete Mathematics (A+), Linear Algebra (A+), Deep Learning (A)

Bachelor of Environmental Engineering & Digital Media Engineering | Catholic University of Korea, Seoul, South Korea (Mar 2013 – Feb 2018)

SKILLS

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, BERT, Transformer, Hugging Face, LLM, RAG, Embeddings, FAISS, BM25, Generative AI, RL (PPO)

Data: pandas, NumPy, SQL, Oracle, MySQL, PostgreSQL, A/B Testing

Backend & DevOps: Spring MVC, NestJS, RESTful API, GraphQL, Docker · Jenkins, GitLab CI/CD, AWS (EC2, S3, RDS), Linux

WORK EXPERIENCE

Jan 2022 – Present **Freelance ML/Software Consultant (Part-time)**

- Provided technical consulting for student research projects and software bug fixes on a project basis

Nov 2020 - Jun 2023 **Konan Technology Inc**, Seoul, South Korea

Software Engineer at AI Solution Development Team

Delivered production search and NLP systems across 4 enterprise clients (Supreme Court, Millie's Library, Heungkuk Insurance, Ministry of SMEs), owning full lifecycle from requirements analysis and infrastructure design through deployment and post-launch stabilization.

- **Supreme Court of Korea** [Java, Spring, Jenkins, GitLab, Python, Linux, Docker, C++, PyTorch, BERT, Oracle]

Built a next-generation electronic litigation search and analysis system, reducing search latency from 5+ minutes to 8 seconds and resolving 375+ misanalysed cases.

- Optimised Java backend algorithms and indexing pipeline to achieve 37x search speed improvement
- Configured Jenkins CI/CD pipelines on GitLab for unit/integration testing and production deployment
- Fine-tuned BERT-based NER model with SFX (dialogue behaviour analysis) and SRL (semantic role labelling) for query intent classification, achieving 4%+ accuracy improvement
- Extracted entities (case numbers, law names, party names) and applied semantic/vector search for query

- expansion and typo correction
- Extracted related/recommended keywords via LDA and time-series analysis of search logs
- Built search analytics dashboards (line, bubble, dendrogram charts) for administrators
- **Millie's Library** [C++, Python, MySQL]
 - Engineered automated indexing system handling 100M+ data points, scalable to 1B+
 - Improved search ranking quality by 38% via custom scoring logic
 - Implemented auto-completion, related-term suggestion, and typo correction
 - Delivered time-series analysis of search terms for recommended/popular keyword features
- **Heungkuk Fire & Marine Insurance** [Java, JavaScript, Oracle]
 - Full-stack development of the web application's search system
 - Implemented AJAX-based continuous scrolling for smooth user experience
 - Built auto-completion and popular search term features
- **Ministry of SMEs and Startups** [Java, PostgreSQL]
 - Developed search engine and RESTful API with text extraction and indexing from AWS S3
 - Implemented recursive file/folder deletion logic for efficient data management

PROJECTS

Feb 2026 - Present **Causal Cue Retrieval and Greedy Route Execution: Mechanistic Analysis of a Transformer Policy in a POMDP T-Maze**

MSc Dissertation, QMUL · Supervisor: Paulo Rauber

- Studied whether a causal Transformer policy can retrieve a no-longer-visible instruction cue from within-episode context, compared against MLP and LSTM baselines trained with PPO in a partially observable MiniGrid environment.
- Designed a controlled ambiguous-memory task with symmetric layout and verified absence of visual leakage; introduced balanced cue success metric min(key, ball) to expose one-sided shortcut policies hidden by aggregate performance.
- Applied retrieval blocking, opposite-memory patching and norm-preserving source exchange: exchanging the real-transition retrieval source reversed both branch decisions while exchanging the padding-derived source reversed neither, though the clean two-sided signature held in only three of six successful checkpoints. Parameter-matched LSTM (h=304, Smin=0.898) also solved the task, shifting the contribution to mechanistic analysis rather than architecture comparison.
- Developed a reusable evaluation framework: branch-level failure decomposition, locked checkpoint selection, and causal context ablation taxonomy, separating genuine memory use from shortcut behaviour.

Feb 2026 – Sep 2026 **Multi-Domain Neural Search Engine and RAG Grounding Evaluation**

MSc Information Retrieval coursework, QMUL · Team project; RAG grounding and statistical layers added independently

- Built a 6-domain search engine: domain-specialised BERT encoders (SciBERT, BioBERT, FinBERT, LegalBERT, ClinicalBERT), a fine-tuned DistilBERT router (3 routing strategies plus an oracle upper bound), hybrid BM25 + FAISS-HNSW retrieval, RRF fusion, ColBERTv2 late-interaction reranking; validated with a 5-step ablation ladder and a hand-annotated cross-domain query set beyond BEIR
- Added a RAG grounding evaluation framework: 6 causal context-perturbation conditions with conditional grounding score, sentence faithfulness and citation support, measuring whether generated answers depend on the retrieved context or on parametric memory across 6 BEIR domains
- Built the A/B testing infrastructure: two-proportion z-test, paired Welch's t-test, Benjamini-Hochberg FDR correction, Cohen's h/d effect sizes across 21 metrics, plus checkpoint/resume, Wilson score CIs, SciFact verification against gold qrels, and small-sample warnings (n<30)

Nov 2025 – Apr 2026 **[GridFlow Trade] AI-Powered BESS Trading Platform (Co-Founder, AI & Backend)**

<https://www.gridflowtrade.com>

- Selected for QMUL QIncubator; pitched to investor panel, Apr 2026

- Engineered 48 domain-driven features (cyclical temporal encodings, price lags 30-min to 7-day, rolling statistics, weather × generation cross-features, LOLP, de-rated margin) on 55,000+ real UK half-hourly settlement periods via proprietary API
- Benchmarked Naive → Linear/Ridge → LightGBM → XGBoost → 2-layer LSTM with time-based train/test split; XGBoost achieved MAE £4.10/MWh ($R^2=0.943$) vs naive baseline £17.13/MWh — 76% error reduction
- Applied ADF stationarity tests, STL seasonal decomposition, ACF/PACF analysis, and leakage-free permutation importance for feature validation
- Built real-time data pipeline integrating NESO & Elexon APIs; deployed XGBoost quantile model to production generating automated charge/discharge/hold signals across 48 daily settlement periods

Apr 2025 **Stock AI YouTube Pipeline** (*Personal Project*)

- Automated weekly AI stock analysis videos using yfinance, OpenAI GPT-4, moviepy, and Stable Diffusion; pipeline fetches data, generates narrated scripts, synthesises visuals, and uploads to YouTube

Feb 2025 – Apr 2025 **BuildU** (*Next.js 15, OpenAI GPT-4, Tesseract.js, Vercel*)

- AI-powered university application advisor: analyses uploaded documents, extracts text via OCR, and generates personalised application strategies
- Integrated OCR via Tesseract.js for document parsing and OpenAI GPT-4 for contextual feedback generation
- Deployed on Vercel with optimised server-side rendering for fast load times

Jul 2023 – Jan 2024 **ATE — Recipe & Ingredients Management** (*React Native/Expo, NestJS, TypeORM, PostgreSQL*)

- Designed the data model and built the backend for a mobile app managing recipes and kitchen inventory, generating a shopping list from the gap between the two
- Implemented backend REST API with NestJS, TypeORM, and PostgreSQL, deployed on AWS EC2

Jul 2022 - Oct 2022 **[Kaggle] RSNA 2022 Cervical Spine Fracture Detection**

- Predicting the probability of a fracture in each of the seven cervical vertebrae from CT scans; built on a public EfficientNetV2 baseline notebook
- EDA, translation and summary of the EfficientNetV2 paper, data preprocessing, model training, and inference

May 2022 - Aug 2022 **[Kaggle] Google AI4Code – Understand Code in Python Notebooks**

- Predicting the next cell to be executed based on the comments and code in Python Jupyter notebooks; built on public baseline notebooks
- Ensemble of CodeBERT and DistilBERT with different weights (0.748 / 0.252) based on the results of running each model

PUBLICATIONS

Oct 2026 Park, H., Kim, E. H., **Choi, W.**, & Zubiaga, A. (2026). *Relevant Is Not Enough: Expert-Audited Decision-Grade Web Evidence for Financial Fact Verification*. In Proceedings of the 11th Workshop on Financial Technology and Natural Language Processing (FinNLP), EMNLP 2026, Budapest. **Accepted (poster), to appear.**

Contribution: annotation quality review on the expert-audited claim–passage set, and pre-submission checks.

ACADEMIC SERVICE

Aug 2026 **Reviewer, FinNLP 2026 @ EMNLP 2026** (*11th Workshop on Financial Technology and Natural Language Processing, ACL SIG-FinTech*)

REFERENCES

Available upon request